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Course: Calculus Beta

Assignment: MyLab Math - Uge 4

Express the following products of complex numbers in the form $a + ib$ where a and b are real numbers.

- a. $(3 + 7i)(2 - 3i)$
 b. $(9 - i)(-2 - 4i)$
 c. $(-1 - 6i)(6 + i)$

a. To multiply two complex numbers, first multiply using the distributive rule.

Multiply using the distributive rule.

$$(3 + 7i)(2 - 3i) = (3)(2) + (3)(-3i) + (7i)(2) + (7i)(-3i)$$

Simplify each product in this expression.

$$(3)(2) + (3)(-3i) + (7i)(2) + (7i)(-3i) = 6 - 9i + 14i - 21i^2$$

The second and third terms in this expression are both multiples of i . Combine these like terms.

$$6 - 9i + 14i - 21i^2 = 6 + 5i - 21i^2$$

The symbol i represents a number whose square equals -1 .

This means that $i^2 = -1$. Substitute this into the last term of the expression from earlier, and then simplify that term.

$$\begin{aligned} 6 + 5i - 21i^2 &= 6 + 5i - 21(-1) && \text{Substitute } i^2 = -1. \\ &= 6 + 5i + 21 && \text{Simplify the last term.} \end{aligned}$$

Simplify this expression by combining like terms.

$$6 + 5i + 21 = 27 + 5i$$

Thus, $(3 + 7i)(2 - 3i) = 27 + 5i$.

b. Start by multiplying using the distributive rule.

$$(9 - i)(-2 - 4i) = (9)(-2) + (9)(-4i) + (-i)(-2) + (-i)(-4i)$$

Simplify each product in this expression.

$$(9)(-2) + (9)(-4i) + (-i)(-2) + (-i)(-4i) = -18 - 36i + 2i + 4i^2$$

The second and third terms in this expression are both multiples of i . Combine these like terms.

$$-18 - 36i + 2i + 4i^2 = -18 - 34i + 4i^2$$

Substitute $i^2 = -1$ into the last term of this expression, and then simplify that term.

$$\begin{aligned} -18 - 34i + 4i^2 &= -18 - 34i + 4(-1) && \text{Substitute } i^2 = -1. \\ &= -18 - 34i - 4 && \text{Simplify the last term.} \end{aligned}$$

Simplify this expression by combining like terms.

$$-18 - 34i - 4 = -22 - 34i$$

Thus, $(9 - i)(-2 - 4i) = -22 - 34i$.

c. Start by multiplying using the distributive rule.

$$(-1 - 6i)(6 + i) = (-1)(6) + (-1)(i) + (-6i)(6) + (-6i)(i)$$

Simplify each product in this expression.

$$(-1)(6) + (-1)(i) + (-6i)(6) + (-6i)(i) = -6 - i - 36i - 6i^2$$

The second and third terms in this expression are both multiples of i . Combine these like terms.

$$-6 - i - 36i - 6i^2 = -6 - 37i - 6i^2$$

Substitute $i^2 = -1$ into the last term of this expression, and then simplify that term.

$$\begin{aligned} -6 - 37i - 6i^2 &= -6 - 37i - 6(-1) && \text{Substitute } i^2 = -1. \\ &= -6 - 37i + 6 && \text{Simplify the last term.} \end{aligned}$$

Simplify this expression by combining like terms.

$$-6 - 37i + 6 = 0 - 37i$$

Thus, $(-1 - 6i)(6 + i) = 0 - 37i$. Since it is not necessary to explicitly include a real part of 0, this product can also be expressed as $-37i$.